Katherine Santiago

UBC Electrical Engineering Student

Vancouver, BC ksantiagoalvarez31@gmail.com (780) 607-0562 katherine-santiago- Portfolio

SKILLS

Hardware: PCB Layout and Fabrication, Analog/digital circuit design, Power Electronics (Buck/Boost Converters, PoE), Zigbee/RF Wireless Systems, Bench Test and Debug (Oscilloscope, Logic Analyzer, DMM), Soldering
Software: C/C++, Python, MATLAB, ESP-IDF, LTSpice, Altium, Ansys Lumerical, SolidWorks, Confluence, JIRA, Google/Microsoft Office Suite
Languages: English (Fluent), Spanish (Fluent), French (Advanced)

EDUCATION

The University of British Columbia, Vancouver, BC Expected May 2027
B.A.Sc in Electrical Engineering
The University of Sydney, Sydney, NSW *(Exchange term)* Aug 2025 - Nov 2025

WORK EXPERIENCE

General Fusion Richmond, BC
Electrical Designer (Part-time Contract) Sep 2026 – Oct 2026

- Tested Arduino Portenta H7 pins to identify usable GPIOs, then measured register turn-on delay with an oscilloscope to assess whether the board could meet system timing requirements.
- Repeated the speed measurement through an electrical-to-LED board, fiber, and an existing custom digitizer board to a laptop, removing the oscilloscope's added delay from the result.
- Began the schematic and PCB layout in Altium for a custom Portenta breakout board intended to replace a 64-channel legacy system.

Electronics and Controls Co-op Jan 2026 – Aug 2026

- Designed a battery-powered Zigbee sensor PCB in Altium over 3 revisions, with dual MAX31856 thermocouple and MAX31865 RTD channels, buck-boost battery regulation with 5V power-mux fallback, switch-configurable inputs, and a custom outline fitting a 3D-printed enclosure.
- Wrote low-power Zigbee end-device firmware (C, ESP-IDF) averaging under 1 mA through sleep-mode power management, and configured/troubleshot the Zigbee2MQTT network; 4 nodes ran deployed for ∼3 weeks on battery, reporting every 30 s.
- Investigated suspected EMI failures on a fielded trigger board: designed, soldered, and bench-tested a circuit-protection revision (TVS diodes, fuse, ferrite filtering) and built an oscilloscope-based EMI test rig to help isolate the root cause.
- Participated in design reviews for the Zigbee and Arduino-based boards and presented the Zigbee project to the entire company (100+ attendees).

HAVEN IAQ Vancouver, BC
Design and Testing Co-op Jan 2025 – Apr 2025

- Developed MATLAB automation scripts to quantify ventilation performance under varying damper and fan configurations, assessing air quality and energy efficiency impacts.
- Calibrated 30+ IoT air quality monitoring devices end-of-line using a vibrating-orifice aerosol generator and aerodynamic particle sizer (APS), and reprocessed multi-month environmental datasets for R&D use.

TECHNICAL PROJECTS

UBC Sailbot – Electrical Communication Systems Member Vancouver, BC
The University of British Columbia Jan 2024 – May 2025

- Designed a custom Altium PCB for the autonomous sailboat's mainframe, featuring a buck converter, switching mechanism, Raspberry Pi header, and Power over Ethernet (PoE) interface.
- Verified RS-232 hat operation on Raspberry Pi and established satellite modem communication via an RS-232-to-inverter interface, validated with an oscilloscope and logic analyzer.

Two-Wheel Self-Balancing Robot (ELEC 391) Vancouver, BC
The University of British Columbia Jun 2025

- Built a self-balancing robot in a team of 3 on an Arduino Nano 33 BLE Sense with a BMI270 IMU, using a Kalman filter for tilt estimation and a PID loop for motor control.
- Contributed to the firmware and PID tuning, achieving stable balance and maneuverability on slopes up to 15°.

OTHER WORK EXPERIENCE AND VOLUNTEERING

UBC Imagine Day Orientations Leader (2023, 2024) • Sales Associate, Dynamite (2024) • Senior Camp Leader, MacDonald Island Park (2023) • Server, Boston Pizza (2022) • Piano Instructor, Discovery Keys Music (2020–2022)